

3.9 Mini File Handling Project: Grade Tracker

گریڈ ٹریکر پہلے 'w' سے فائل بناتا ہے، پھر 'a' سے نیا طالب علم جوڑتا ہے، پھر 'r' سے تمام سطریں پڑھ کر اوسط نکالتا ہے۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 17 of 17

3.9 Mini File Handling Project: Grade Tracker

LEARN IT

This project uses 'w' to create a grades file, 'a' to add a student later, and 'r' to read every line and compute an average. It combines file handling, string methods and arithmetic.

Write names and marks ('w') → Append a new student ('a') → Read lines ('r') → Split, total, average

```
with open("grades.txt", "w") as f:
    f.write("Alice: 88\n")
    f.write("Bob: 75\n")
    f.write("Carol: 92\n")
    print("Grades file created.")

with open("grades.txt", "a") as f:
    f.write("David: 81\n")
    print("New student appended.")

with open("grades.txt", "r") as f:
    lines = f.readlines()
    total = 0
    for line in lines:
        name, grade = line.strip().split(":")
        print(f"{name} scored {grade}")
        total += int(grade)
    average = total / len(lines)
    print(f"\nClass average: {average:.1f}")
```

WORKED EXAMPLE

What happens in memory and on disk.

1. 'w' creates grades.txt (or wipes an old one) with three lines.
2. 'a' adds David: 81\n at the end. Alice, Bob and Carol stay.
3. .readlines() returns
['Alice: 88\n', 'Bob: 75\n', 'Carol: 92\n', 'David: 81\n'].
4. .strip() removes the trailing \n. .split(":") turns "Alice: 88" into ["Alice", " 88"].
5. int(grade) accepts the extra space: int(" 88") is 88.
6. Total = 88 + 75 + 92 + 81 = 336. Average = 336 / 4 = 84.0. The format :.1f prints one decimal place.

OUTPUT

```
Grades file created.
New student appended.
Alice scored 88
Bob scored 75
Carol scored 92
David scored 81

Class average: 84.0
```

The extra space before each mark in the output comes from the space after the colon in the file ("Alice: 88"). int() ignores it. To tidy the printout you could use grade.strip() as well; the average would not change.

REMEMBER IT

Same file, three modes, three `with` blocks. Never open the same file in `'w'` when you only meant to add David — that would delete Alice, Bob and Carol.

COMMON MISTAKE

If a line has no colon, `split(":")` will not produce two parts and the unpacking `name, grade = ...` will raise `ValueError`. Keep the `Name: mark` format. An empty file would make `len(lines)` zero and then `total / 0` would fail — a real system should check that, just as `divide` checked for zero.

اردو میں سمجھیں

گریڈ ٹریکر پہلے `'w'` سے فائل بناتا ہے، پھر `'a'` سے نیا طالب علم جوڑتا ہے، پھر `'r'` سے تمام سطریں پڑھ کر اوسط نکالتا ہے۔ ہر سطر کو `strip` اور `split(":")` سے نام اور نمبر میں توڑا جاتا ہے۔ اوسط = کل نمبر ÷ طلبہ کی تعداد۔ `'w'` دوبارہ استعمال کرنے سے پرانے ریکارڈ مٹ جائیں گے۔

CHECK YOURSELF

1. What is the class average in the sample, and how is it calculated?
2. Why is David written with mode `"a"` rather than `"w"`?

Answers: 1. 84.0 because $(88+75+92+81)/4 = 336/4 = 84.0$. 2. Because `"a"` appends; `"w"` would erase the three existing students.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. `open(path, "r")` is for:

- a. Reading
- b. Wiping the file
- c. Appending only
- d. Creating a DB

MCQ 2. Mode "w":

- a. Write (truncates existing)
- b. Read only
- c. Append
- d. Binary XOR

MCQ 3. Mode "a":

- a. Append at the end
- b. Overwrite from start
- c. Read
- d. Delete the file

MCQ 4. You should:

- a. `close()` (or use `with`)
- b. Leave files locked
- c. Never flush
- d. Always use `w` for logs

MCQ 5. `read()` returns:

- a. The whole content as `str` (text mode)
- b. A dict always
- c. A pandas DataFrame
- d. A stack

MCQ 6. This is ★ because practicals ask you to:

- a. Write a small file program
- b. Draw only ER
- c. Only HCI
- d. Only OSI

Extra short questions

1. The three text modes and a danger of `w`.
2. `write` vs `writelines`.
3. Why `close`?
4. A `FileNotFoundError` happens when:

Extra long question

1. Write a program that appends a line to `marks.txt` and then reads all lines. Show the file after two runs. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. `r` read, `w` write/overwrite, `a` append. `w` erases previous content.
2. `write` a string; `writelines` a sequence of strings.
3. Flush + unlock. Crash before `close` can lose data.
4. `r` on a path that does not exist.

Long

1. open `a`; write; close; open `r`; print. Second run has two lines. Warn that `w` would have kept only the last.